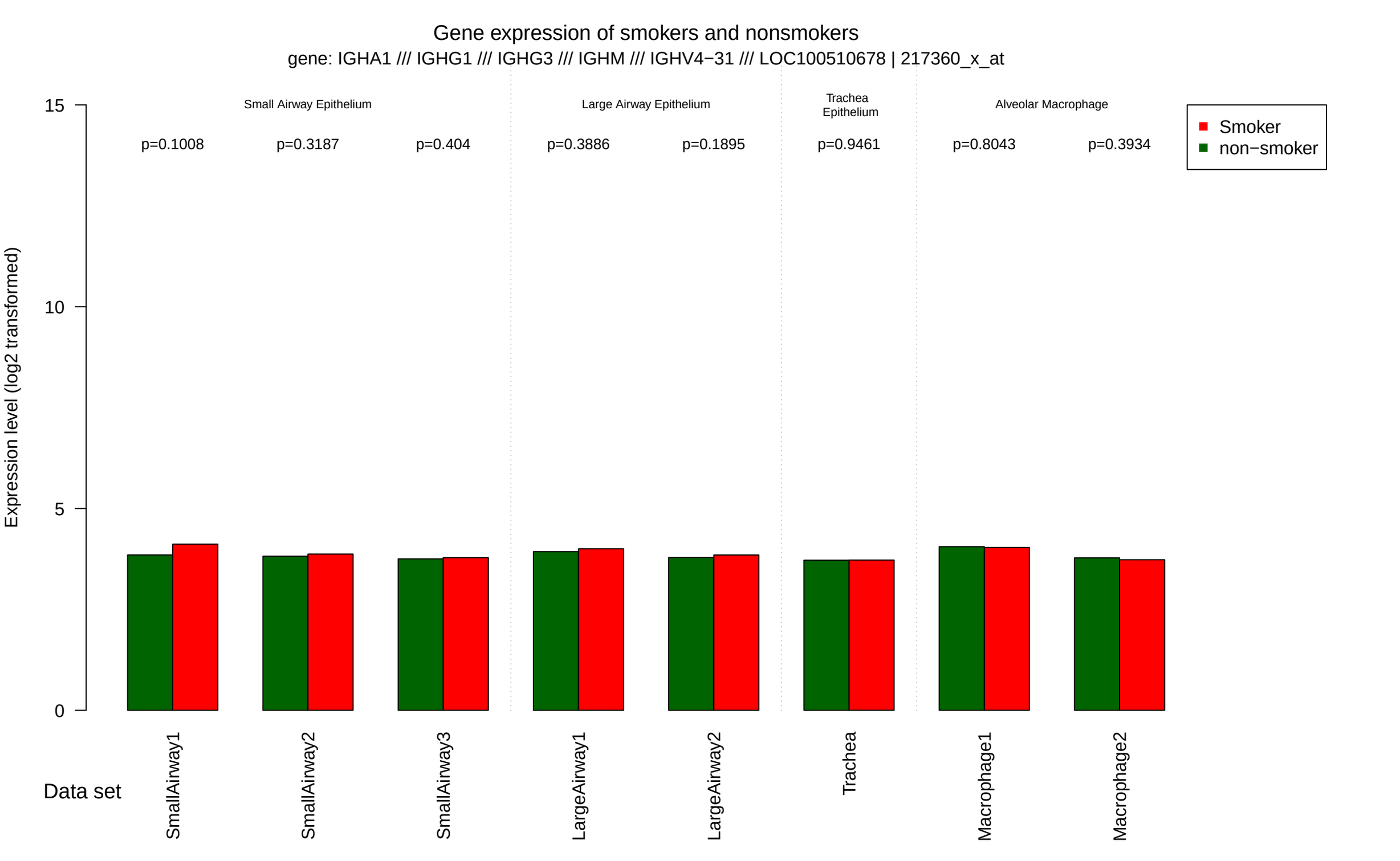
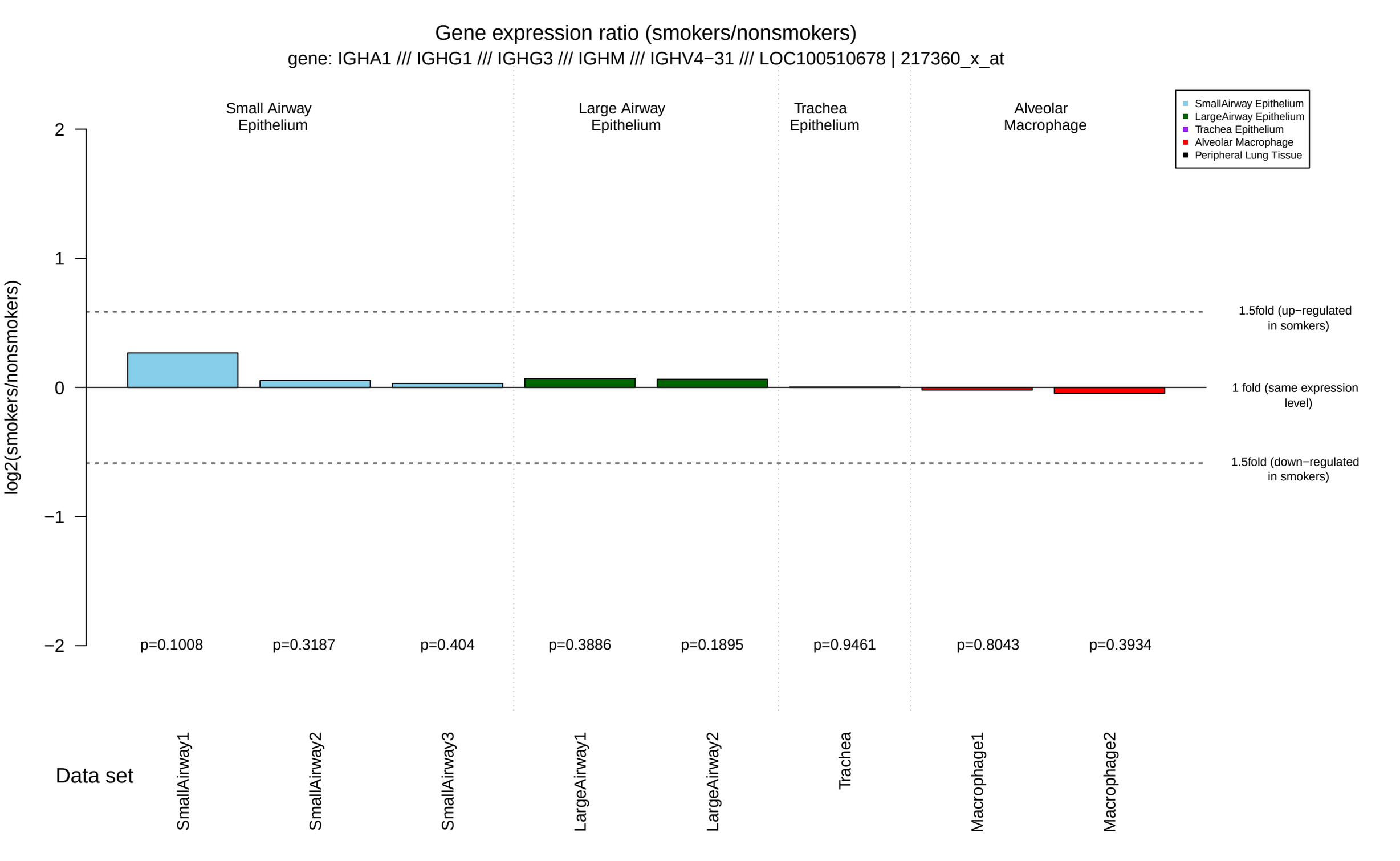
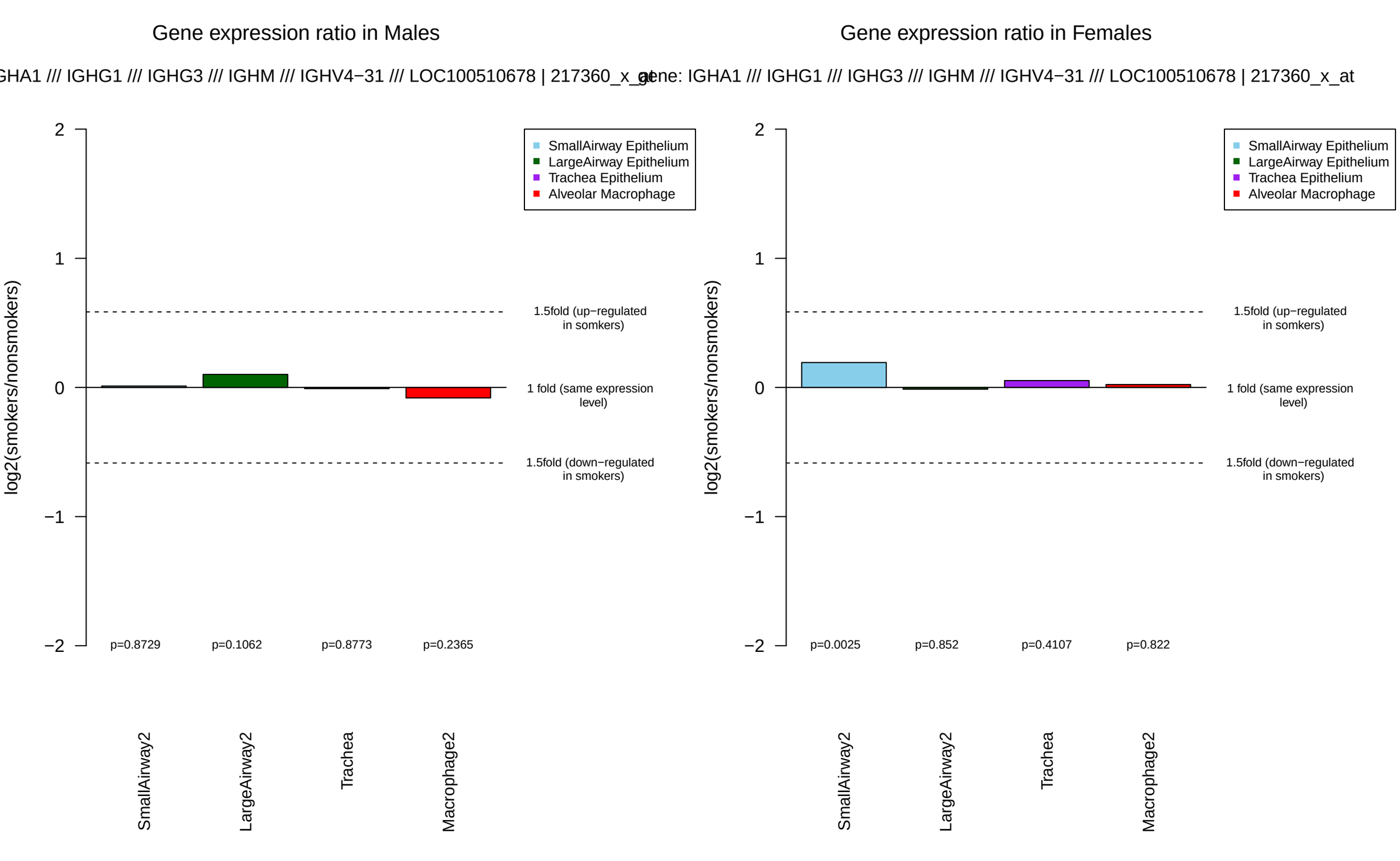
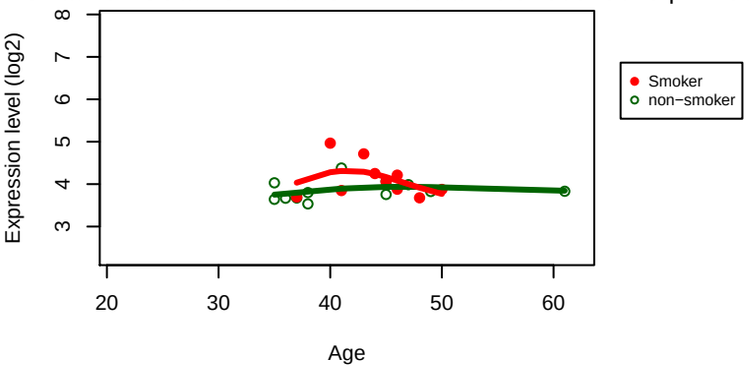

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 217360_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1008
p=0.3187
p=0.404
p=0.3886
p=0.1895
p=0.9461
p=0.8043
p=0.3934
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

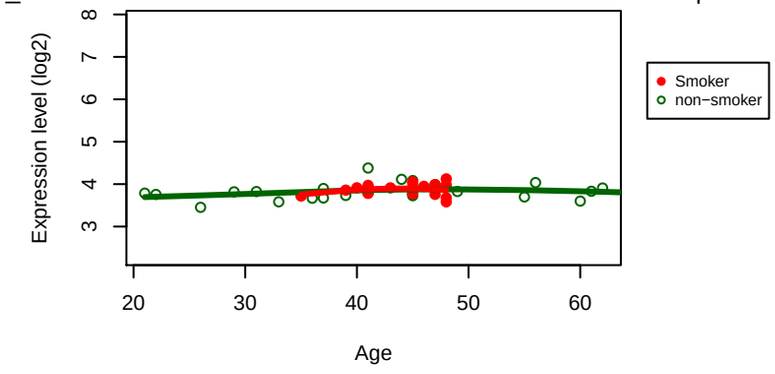

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 217360_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1008
p=0.3187
p=0.404
p=0.3886
p=0.1895
p=0.9461
p=0.8043
p=0.3934
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio in Males
GHA1 /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 217360_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8729
p=0.1062
p=0.8773
p=0.2365
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
Gene: IGHA1 /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 217360_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0025
p=0.852
p=0.4107
p=0.822
SmallAirway2
LargeAirway2
Trachea
Macrophage2

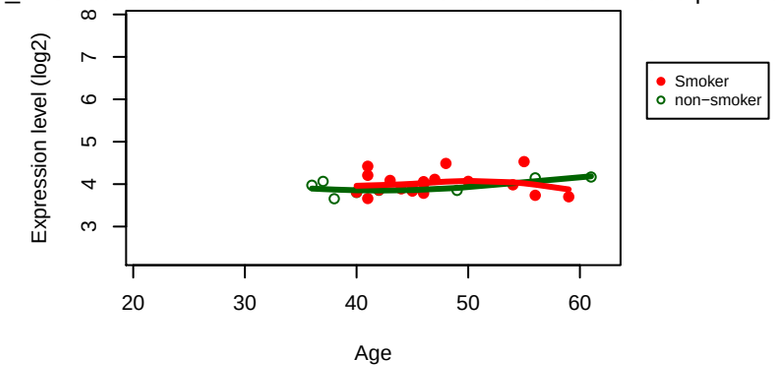
set: SmallAirway1 (p=0.1008)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 217360_x_a



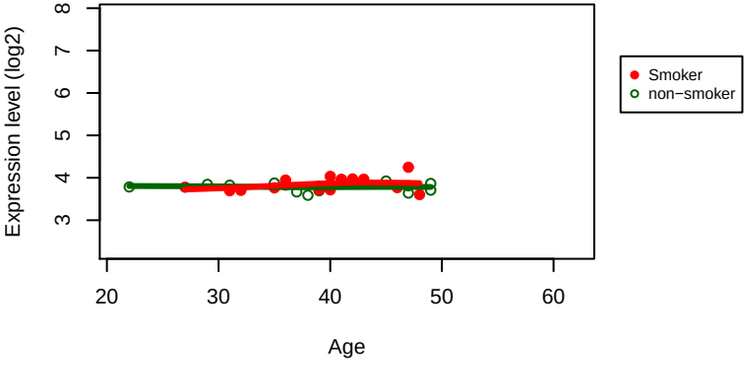
set: SmallAirway2 (p=0.3187)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 217360_x_a



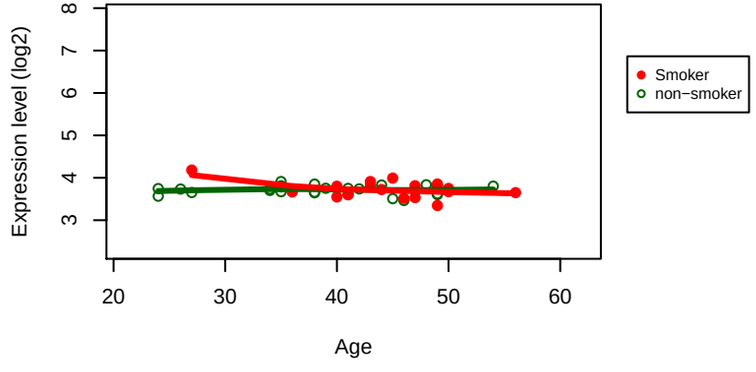
set: LargeAirway1 (p=0.3886)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 217360_x_a



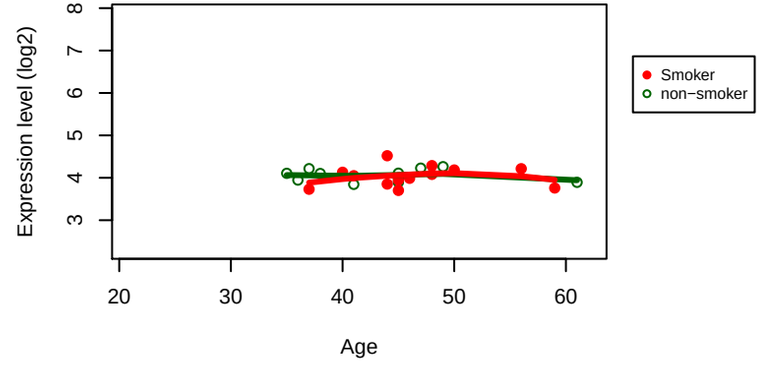
set: LargeAirway2 (p=0.1895)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 217360_x_a



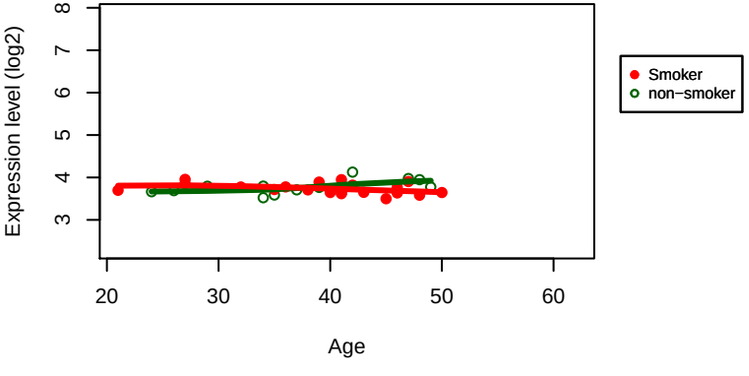
set: Trachea (p=0.9461)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 217360_x_a

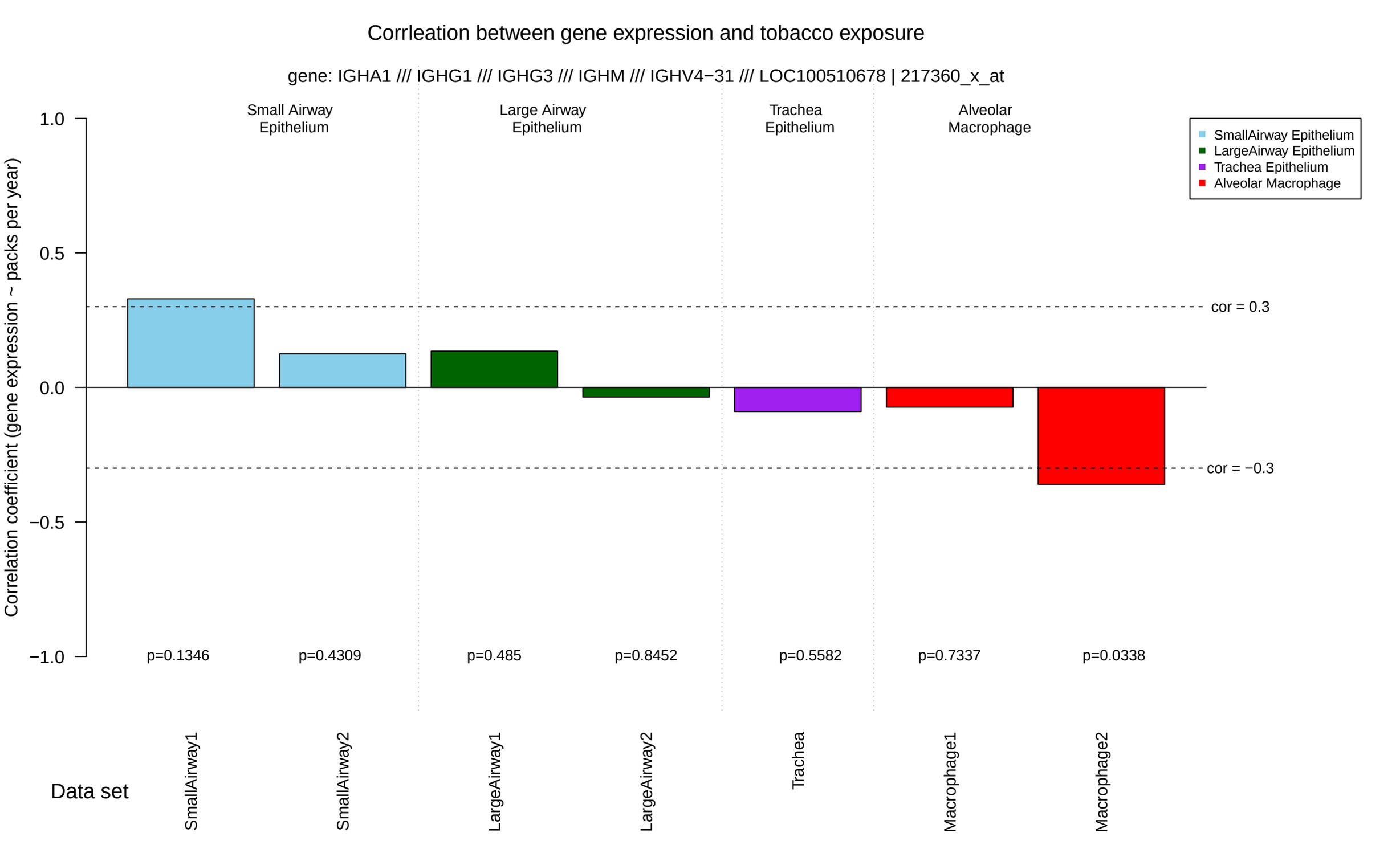


set: Macrophage1 (p=0.8043)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 217360_x_a



set: Macrophage2 (p=0.3934)
HG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 217360_x_a




Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 217360_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1346
p=0.4309
p=0.485
p=0.8452
p=0.5582
p=0.7337
p=0.0338
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2